

Data Flow Architecture

```
graph TD
    Yahoo_Finance[Yahoo Finance] --> Stock_ETL_Service[Stock ETL Service]
    Stock_ETL_Service --> stock_prices[(stock_prices  
(PostgreSQL))]
    Alpha_Vantage[Alpha Vantage] --> News_ETL_Service[News ETL Service]
    News_ETL_Service --> news_metadata[(news_metadata  
(PostgreSQL))]
    news_metadata --> Sentiment_Service[Sentiment Service]
    Sentiment_Service --> sentiment_scores[sentiment_scores]
    stock_prices --> Research_Service[Research Service]
    news_metadata --> Research_Service
    sentiment_scores --> Research_Service
    Research_Service --> risk_scores[risk_scores]
```

Yahoo Finance
|
v
Stock ETL Service
|
v
stock_prices
(PostgreSQL)

Alpha Vantage
|
v
News ETL Service
|
v
news_metadata
(PostgreSQL)

news_metadata
|
v
Sentiment Service
|
v
sentiment_scores

stock_prices
news_metadata
sentiment_scores
|
v
Research Service
|
v
research_scores

research_scores
sentiment_scores
|
v
Risk Service
|
v
risk_scores